

Introduction

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This introductory chapter outlines the major focus, specific tools, and strategies of computational neuroscience. We argue that theoretical and computational studies are important in order to comprehend and guide experimental investigation. The nature and role of models in research is discussed. A model is an abstraction of the real systems used to investigate specific questions, and this chapter outlines the guiding principles for models used throughout the book. In addition, a high-level theory of how the brain works is included in this chapter to guide some discussions in the following chapters. The last section of this chapter includes a short guide to the book, outlining the path we are taking while embarking on a fascinating scientific venture.

1.1 What is computational neuroscience?	1
1.2 What is a model?	5
1.3 Is there a brain theory?	8
1.4 A computational theory of the brain	13
Exercises	16
Further reading	17

1.1 What is computational neuroscience?

The scientific area now commonly called computational neuroscience uses distinct techniques and asks specific questions aimed at advancing our understanding of the nervous system. A brief definition might be:

Computational neuroscience is the theoretical study of the brain used to uncover the principles and mechanisms that guide the development, organization, information-processing and mental abilities of the nervous system.

Thus, computational neuroscience is a specialization within neuroscience. Neuroscience itself is a scientific area with many different aspects. Its aim, and therefore that of computational neuroscience, is to understand the nervous system, in particular the part concentrated in the skull, that we call the brain. The brain is studied by researchers who belong to diverse disciplines, including physiology, psychology, medicine, computer science and mathematics, to name but a few. Neuroscience emerged from the realization that interdisciplinary studies are vital to further our understanding of the brain. While huge progress in our understanding of brain functions has been made in the past few decades, there are many questions that we can only tackle by combined efforts: How does the brain work? What are the biological mechanisms involved? How is it organized? What are the information-processing principles used to solve complex tasks such as perception? How did it evolve? How does it change during the lifetime of organisms? What is the effect of damage to particular areas and the possibilities of rehabilitation? What are the origins of degenerative diseases and possible treatments? These are questions asked by neuroscientists in many different sub-fields, using a multitude of different research techniques.

1.1.1 Tools and specializations in neuroscience

Many techniques are employed within neuroscience to answer those important questions about brain operations. These include genetic manipulations; recording of cell activities in cultured cells, brain slices, or even whole brains; optical imaging; non-invasive functional brain imaging; psychophysical measurements; and computational simulations. Each of these techniques is complicated and laborious enough to justify a specialization of neuroscientists in particular techniques. Therefore, we speak of neurophysiologists, cognitive scientists, and anatomists. It is, however, vital for any neuroscientist to develop a basic knowledge of all major techniques, so he or she can comprehend and utilize the contributions made in other specializations. The significance of any technique has to be evaluated in view of its specific problems and limitations, as well as the specific aim of the technique. Computational neuroscience is one of the areas in neuroscience with increasing importance, and a basic comprehension of this field is becoming essential for all neuroscientists.

Computational and theoretical neuroscience develops and tests hypotheses of the functional mechanisms of the brain. A major focus in this area is therefore the development and evaluation of *models*. Computational neuroscience can be viewed as a specialization within theoretical neuroscience that employs computers to simulate models. The major reason for using computers is that the complexity of models in this area is often beyond analytical tractability. For such models we have to employ carefully designed numerical experiments to be able to compare the models to experimental data. However, we do not want to restrict our studies to this tool, because analytical studies can often give us a much deeper and more controlled insight into the features of models and the reasons behind numerical findings. Whenever possible, we try to include analytical as well as numerical techniques, and we will use the term computational neuroscience synonymously with theoretical neuroscience. The term 'computational' also stresses that we are interested, in particular, in the computational and information-processing aspects of brain functions.

Although computational neuroscience is theoretical by its very nature, it is important to bear in mind that models must be gauged on experimental data; they are otherwise useless for understanding the brain. Only experimental measurements of the real brain can verify 'what' the brain actually does. In contrast to the experimental domain, computational neuroscience tries to speculate 'how' the brain operates. Such speculations are developed into hypotheses, realized into models, evaluated analytically or numerically, and tested against experimental data. Also, models can often be used to make further predictions about the underlying phenomena.

Ideally, we want to integrate experimental facts from different levels of investigation into a coherent model of how the brain works. The interaction between different disciplines, computational studies and experiments, is illustrated in Fig. 1.1. Important experimental input to computational neuroscience comes from neuroanatomy, such as the morphology and functional connectivity of brain structures; from neurophysiology, in which the behaviour of single neurons is investigated; and from psychology, in which behavioural effects are studied with psychophysical experiments. Computational neuroscientists

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1.1.2 Levels of investigation

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use mathematical models for the description of experimental facts, borrowing methods from a wide variety of disciplines, such as mathematics, physics, computer science and statistics. Investigations of hypotheses with the aid of models leads to specific predictions that have to be verified experimentally. The comparison of model predictions with experimental data can then be used to refine the hypotheses and to develop more accurate models, or even models that can shed light on different phenomena. The studies within computational neuroscience can also help to develop applications such as advanced analysis of brain-imaging data, technical applications that utilize brain-like computations, and ultimately, better treatment of patients with brain damage and other brain-related disorders.

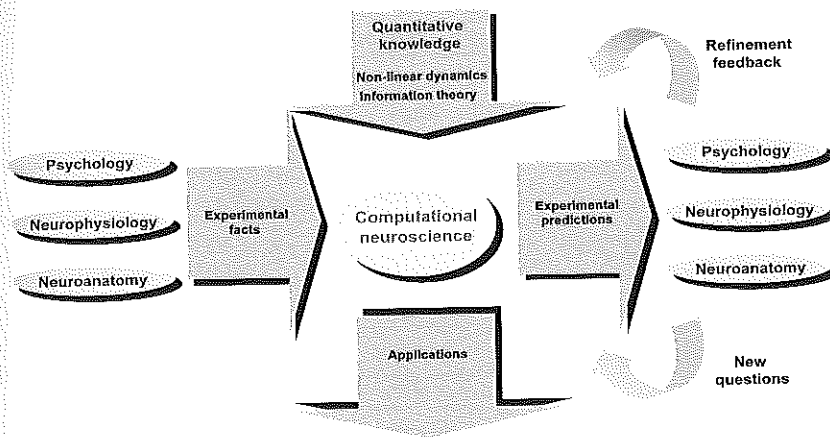


Fig. 1.1 Illustration of the role of computational neuroscience in the integration of experimental facts from different levels of investigation. The models developed in computational neuroscience make predictions that can be verified experimentally. The close comparison of experiments with model predictions can then be used to make refinements to the models (or may even lead to the development of new approaches) that can further our understanding of brain systems and can make new predictions that have to be verified experimentally. Also, such research frequently leads to new applications, such as advanced analysis techniques or new patient treatment techniques. [Adapted from Gustavo Deco, personal communication.]

1.1.2 Levels of organization in the brain

The nervous system has many levels of organization on spatial scales ranging from the molecular level of a few Angstrom ($1\text{\AA} = 10^{-10}\text{m}$), to the whole nervous system on the scale of over a metre. Biological mechanisms at all these levels are important for the brain to function. Different levels of organization in the nervous system are illustrated in Fig. 1.2. An important structure in the nervous system is the *neuron*, which is a cell that is specialized for signal processing. Depending on external conditions, neurons are able to generate electric potentials that are used to transmit information to other cells to which they are connected. Mechanisms on a subcellular level are important for such information-processing capabilities. Neurons use cascades of biochemical reactions that have to be understood on a molecular level. These include, for example, the transcription of genetic information which influences information-processing in the nervous system. Many structures within neurons can be identified with specific functions. For example, *mitochondria* are structures important for the energy supply in the cell, and *synapses* mediate information transmission between cells. The complexity of a single neuron,

and even isolated subcellular mechanisms, makes computational studies essential for the development and verification of hypotheses. It is possible today to simulate morphologically reconstructed neurons in great detail, and there has been much progress in understanding important mechanisms on this level.

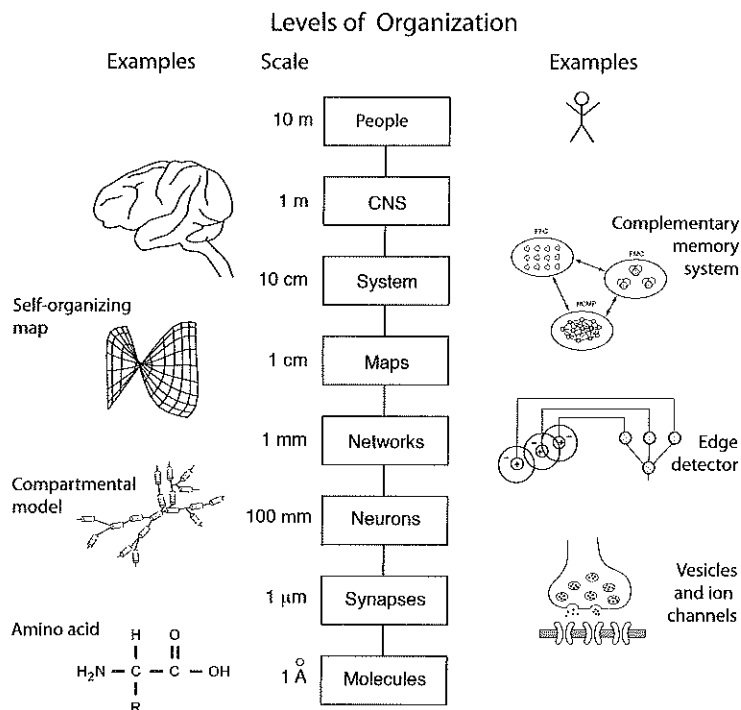


Fig. 1.2 Some levels of organization in the central nervous system on different scales [adapted from Churchland and Sejnowski, *The computational brain*, MIT Press (1992)].

However, single neurons certainly do not tell the whole story. Neurons contact each other and thereby compose *networks*. A small number of interconnected neurons can exhibit complex behaviour and enable information-processing capabilities not present in a single neuron. Understanding networks of interacting neurons is a major domain in computational neuroscience. Networks have additional information-processing capabilities beyond that of single neurons, such as representing information in a distributed way. An example of a basic network is the *edge detector* formed from a centre-surround neuron as proposed by *Hubble* and *Wiesel*. The illustrated levels above the one labelled 'Networks' in Fig. 1.2 are also composed of networks, yet with increasing size and complexity. An example on the level termed 'Maps' in Fig. 1.2 is a *self-organizing topographic map*, which is part of an important discussion in this book.

The organization does not stop at the map level. Networks with a specific architecture and specialized information-processing capabilities are composed into larger structures that are able to perform even more complex information-processing tasks. System-level models are important in understanding higher-order brain functions. The central nervous system depends strongly on the

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dynamic interaction of many specialized subsystems, and the interaction of the brain with the environment. Indeed, we will see later that active environmental interactions are essential for brain development and function.

Although an individual researcher typically specializes in mechanisms of a certain scale, it is important for all neuroscientists to develop a basic understanding of the functionalities of different scales in the brain. Computational neuroscience can help the investigations at all levels of description, and it is not surprising that computational neuroscientists investigate different types of models at different levels of description. Computational methods have long contributed to cellular neuroscience, and computational cognitive neuroscience is now a rapidly emerging field. The contributions of computational neuroscience are, in particular, important to understand non-linear interactions of subprocesses. Furthermore, it is important to comprehend the interactions between different levels of description, and computational methods have proven very useful in bridging the gap between physiological measurements and behavioural correlates.

1.2 What is a model?

Modelling is an integral part of many scientific disciplines, and neuroscience is no exception. The more complex a system is, the more we have to make simplifications and build example systems to provide insights into aspects of the complex system under investigation. The term 'model' appears frequently in many scientific papers, and describes a vast variety of constructs. Some papers present a single formula as a model, some papers fill several pages with computer code, and some describe with words a hypothetical system. We need to understand what a model is, and, in particular, what the purposes of models are.

It is important to distinguish a model from a hypothesis or a theory. Scientists develop hypotheses of the underlying mechanisms of a system that have to be tested against reality. In order to test a specific feature of a hypothesis, we build a model that can be evaluated. Sometimes we try to mimic real systems under artificial means, in order to be able to test the systems by different conditions, or to make measurements that would not be possible in a 'real' system. A model is hence a simplification of a system in order to test particular aspects of the system or hypothesis. A brief explanation of a model, which is useful to remember throughout this book and in research, is:

A model is an abstraction of a real-world system to demonstrate particular features of, or investigate specific questions about, the system. Or, in more scientific terms, a model is a quantification of a hypothesis to investigate the hypothesis.

A good example of this is the use of models in the field of architecture. Small-scale paper models of buildings, or computer graphics (Fig. 1.3) generated with sophisticated three-dimensional graphics packages, can be used to get a first impression of the physical appearance and aesthetic composition of a design. A model has a particular purpose attached to it and has to be viewed in this light.

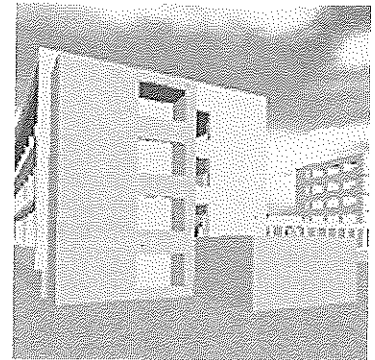


Fig. 1.3 A computer model of a building which gives a three-dimensional impression of the design.

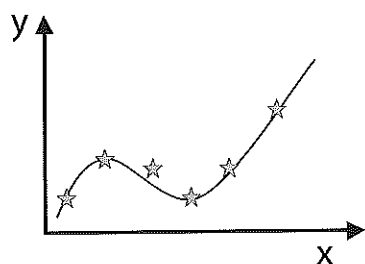


Fig. 1.4 The graph shows some data points, plotted with star symbols, such as data derived from experimental measurements. Also shown is a model represented by the line, and this curve fits the data reasonable well. The curve can be derived from a simple mathematical formula that fits the data points (phenomenological model), or result from more detailed models of the underlying system.

A model is not a recreation of the 'real' thing. The paper model of a house cannot be used to test the stability of the construction, a purpose for which a building engineer uses different models. In such models, it is important to scale down physical properties of the building materials regardless of the physical appearance, such as the colour of the building.

1.2.1 Phenomenological and explanatory models

In science, we typically represent experimental data in the form of graphs, and then seek to describe these data points with mathematical functions (Fig. 1.4). An example of this is the 'modelling' of response properties (tuning curves) of neurons in the *lateral geniculate nucleus* (LGN), which can be fitted with a specific class of functions called *Gabor functions* by adjusting the parameters of these functions. Gabor functions are therefore said to be 'models' of the receptive fields in the LGN. Of course, this *phenomenological model* does not tell us anything about the biophysical mechanisms underlying the formation of receptive fields and why cells respond in this particular way, so such a 'model' seems rather limited. Nevertheless, it can be useful to have a functional description of the response properties of LGN cells. Such parametric models are a shorthand description of experimental data that can be used in various ways. For example, if we want to study a model of the primary visual cortex, to which these cells project, then it is much easier to use the parametric form of LGN responses as input to the cortical model, rather than including further complicated models of the earlier visual pathway in detail.

As scientists, we want to find the roots of natural phenomena. The explanations we are seeking are usually deeper than merely parameterizing experimental data with specific functions. Most of the models in this book are intended to capture processes that are thought of as being the basis of the information-processing capabilities of the brain. This includes models of single neurons, networks of neurons, and specific architectures capturing brain organizations. Models are used to study specific aspects of a hypothesis or theory, but can also help to interpret experimental data.

A major role of models in science is to illustrate principles underlying natural phenomena on a conceptual level. Sometimes the level of simplification and abstraction is so crude that scientist talk about *toy models*, although this terminology undermines slightly the importance of such models in science. The simplifications made in such models might be necessary to employ analytical methods to analyse such models at a depth that is not possible in more realistic models. The educational importance of these toy models should not be underestimated, in particular, in demonstrating principal mechanisms of natural phenomena. It is easy to complicate things, but the real scientific challenge is to simplify theoretical concepts.

The current state of neuroscience, often still exploratory in nature, frequently makes it difficult to find the right level of abstraction to properly investigate hypotheses. Some models in computational neuroscience have certainly been too abstract to justify claims derived from them. On the other hand, there is a great danger in keeping too many details that are not essential for the scientific argument. Models are intended to simplify experimental data, and thereby to

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identify which details of the biology are essential to explain particular aspects of a system. Modelling the brain is not a contest in recreating the brain in all its details on a computer. It is questionable how much more we would comprehend the functionality of the brain with such complex and detailed models. Models must be carefully constructed, and reasonable simplification are the result of careful scientific investigations and insight into natural processes. It is not always possible to justify all assumptions adequately, but it is important to at least clarify which assumptions have been made and to argue about them. The purpose of models is to better comprehend the functionality of complex systems, and simplicity should be a major guide in designing appropriate models. This philosophy is sometimes called the *principle of parsimony*, also known as *Occam's razor*. Basically, we want the model as simple as possible, while still capturing the main aspects of data that the model should capture. We will see this principle in action throughout this book, and discuss the issue specifically in conjunction with machine learning in Chapter 6.

1.2.2 Models in computational neuroscience

Where do we start when trying to explain brain functions? Do we have to re-build the brain in its entirety with computational techniques in order to understand it? Currently, many neuroscientists are collaborating in an ambitious project, called the *Blue Brain Project*, to implement in a computer, with as much detail as possible, neocortical circuits. This type of detailed project can show us which emergent phenomena can be expected in the brain, or where gaps in our knowledge require further investigation. However, even if it was possible to simulate a whole brain on a computer, with all the details from biochemistry to large-scale organization, this would not explicitly mean that we have better explanations of brain functions. What we are looking for, at least in this book, is a better comprehension of brain mechanisms on explanatory levels. It is therefore important to learn about the art of *abstraction*, making suitable simplifications to a system without abolishing the important features we want to comprehend.

The level that is most appropriate for the investigation and explanatory abstraction depends on the scientific question. For example, epileptic seizures are known to be caused by synchronization of whole brain areas, and an understanding of brain dynamics on a systems level is important to comprehend such disorders. This should, of course, not exclude the search for causes on much smaller scales. We know that Parkinson's disease is caused by the death of dopaminergic neurons in the *substantia nigra*, and the role of dopamine in the initiation of motor actions is important to comprehend the full scale of impairments, and to develop better methods of coping with such conditions. The causes of the cell death of dopaminergic neurons are still not known, and it is important to find the reasons, possibly on a genetic level in this case, to enable a real treatment of Parkinson's disease. Therefore, various levels must be investigated, and we have to learn to make connections between the different levels in order to follow how the low-level circumstances, such as mechanisms on a genetic level or biochemical processes in neurons, can influence the characteristics of large-scale systems, such as behaviour of an organism.

1.3 Is there a brain theory?

It is often said that the brain is the most complex system that we know of in nature, and understanding how it works is a large, if not impossible, task. It is true that understanding how the brain works seems difficult when trying to reverse engineer it. However, reverse engineering a system can even be difficult for systems that we created in the first place. For example, imagine measuring the varying states of a transistor in a computer while the computer is running a word processor. Such measurements give important clues from which it should be possible to discover regularities when certain operations are repeated on the computer. Also, such measurements make it possible to discover important principles that must be at work in the digital computer system, such as the discrete nature of information representations. However, it seems a daunting task to recreate a computer program such as a word processor from these data, even if we were able to analyse a large number of measurements of many parts of the computer at the same time. The direct reverse engineering of the brain from data seems even more challenging, given the biological nature of the object. However, we can use the data to understand the principles of brain processing, and we can then use this knowledge to build brain-style information-processing systems. This approach is taken in this book.

Correspondingly, there has been some shift in the research approach of the neuroscience community. The past decades have been an era in neuroscience marked by a flood of explorations. Recordings with micro-electrodes from single cells contributed significantly to this exploration, and searching for response properties of neurons is at least as exciting as it must have been for explorers such as Marco Polo to discover new lands. New brain-imaging techniques, such as functional magnetic resonance imaging (fMRI), make it possible to monitor living brains of subjects performing specific mental tasks. Explorations of brain functions with such techniques have been essential in advancing our knowledge in neuroscience. However, while a mountain of data has been gathered for brain functions, using a multitude of techniques, we are now slowly entering a new phase in neuroscience, that of formulating more quantitative hypotheses of brain functions. This shift in the focus of neuroscience research demands some more specific experimental analysis and more dedicated tests of such hypothesis. It is increasingly important to formulate a quantitative hypothesis, and possible alternatives, in such a way that the hypothesis can be tested experimentally.

This new era of neuroscience sounds a lot like quantitative scientific areas such as chemistry or physics, and while a quantitative analysis will generate new breakthroughs in our understanding of brain functions, the ultimate question is if there can be a *brain theory*. We could take the position that in order to understand the brain we need to understand all of the structural details and the current state of a particular brain. This is no different in other scientific areas such as physics. For example, to completely describe the physics of an individual aeroplane we have to know the precise location and form of each nut and bolt and all other structural details up to the amount of dirt on the wings and the details of the air it is flying through. Another example is that of a pot of boiling water, which consists of a lot of individual molecules in a very

dynamic state. It seems impossible to know the precise location and form of each molecule, but is the boiling water a complex system? Important questions arise. What was the reaction of a pot of boiling water to the dynamics of the flowing air? We know today that the interactions between the water molecules and the air molecules are more than enough to cause the boiling.

There is a similar scientific question in neuroscience. It is certainly true that the brain is a very complex system. It stores memories, makes decisions, and is involved in all sorts of associations. It is why the environment affects the brain. Theories of brain function are needed to understand the brain. We have already achieved a lot of progress in this goal of this book.

1.3.1 Brain as a computer

Standard computers consist of processors, memory, and input/output devices. Microprocessors, registers, and buses are the basic components which are used to process data and execute instructions. Solving a task is done by precisely allocating resources to the problem. The programmer writes a program to solve the problem.

In contrast to a standard computer, the brain has many respects in which it is different. There are lots of theories of the brain, and they are mainly used to stress this difference.

dynamic state. Measuring all the microscopic details in the last two examples seems impossible. Yet, we have a fairly good understanding of the process of boiling water and why an aeroplane can fly. This was not possible overnight but is the result of dedicated scientific research over the past few centuries. Important for the success in physics, chemistry, and other scientific disciplines, was the realization of the right level of description or the right level of abstraction of a problem. For example, we learned to describe the average behaviour of the molecules in an ideal gas, which gave us the fundamentals of thermodynamics and ultimately led to a better understanding of the mechanisms of flowing air that enables engineers to construct more efficient aeroplane. We know today about the essential quantum nature of atomic and subatomic interactions, but a description of Mount Everest on this level is not reasonable. There are geological theories of mountain formation that are more appropriate than employing quantum theory to these questions.

There is no reason to conjecture that the brain cannot be tractable with similar scientific rigour to that developed in other disciplines. The brain is certainly more complex than a gas of weakly interacting atoms. However, there are very fundamental questions that we can attack; for example, how a network stores memories that can be recalled in an associative way. Indeed, we have made considerable progress with this question, considering our understanding of associative networks discussed in this book. Another fundamental question is why the brain is relatively stable, while still being able to adapt to novel environments. Brain theories of this kind are now emerging, and some of these theories will be discussed in this book. It may be too early to talk about a single brain theory, but there is no reason to suspect that theories will advance our understanding of brain functions. Indeed, I think that major breakthroughs have already been made, which need to be brought to a wider audience. The goal of this book is to contribute to this endeavour.

1.3.1 Emergence and adaptation

Standard computers, such as PCs and workstations, have one or more central processors. Each processor is rather complex, with specialized hardware and microprograms implementing a variety of functions, such as loading data into registers, adding, multiplying and comparing data, as well as communicating with external devices. These basic functions can be executed by instructions, which are binary data loaded into a special interpreter module. Complicated data processing can be achieved by writing, often lengthy, programs, which are instructions representing a collection of the basic processor functions. When solving a task with a computer, we have to instruct the machine to follow precisely all the steps that we determined beforehand would solve a particular problem. The sophistication of computers basically reflects the ingenuity of the programmer.

In contrast, information-processing in the brain is very different in several respects. The brain employs simpler processing elements than computers, but lots of them. To explore information processing in networks of neurons, we will mainly use very simple abstractions of real neurons, which we call *nodes* to stress this drastic simplification. These fundamental processing units can be

implemented in hardware, or simulated on a standard computer; for the discussions in this book this does not make a difference. We keep the functionality of nodes as simple as possible for the sake of employing lots of them, typically hundreds, thousands, or even more. The usage of many parallel working processors has motivated the term *parallel distributed processing* in this area. However, with this term one is tempted to think that the processes are independent because only processes that are independent can be processed on different processors in parallel. In contrast to this, a major ingredient of information-processing in the brain is the *interaction of neurons*, and the interaction of neurons is accomplished by assembling them into large networks.

It is the interaction of nodes that enables processing abilities not present in single nodes. Such capabilities are good examples of emergent properties in rule-based systems. *Emergence* is the single most defining property of neural computation, distinguishing it from parallel computing in classical computer science, which is mainly designed to speed up processing by distributing independent algorithmic threads. Interacting systems can have unique properties beyond the mere multiplication of single processor capabilities. It is these types of abilities we want to explore and utilize with neural networks. These system properties are labelled as emergent to stress that we did not encode these properties directly into the system. To better appreciate this, we distinguish the description of a system on two levels, the level of basic rules defining the system, and the level of description aimed at understanding the consequences of such rules. In the study of neural networks we are interested in understanding the consequences of interacting nodes.

Scientific explanations were dominated in the past by the formulation of a set of principles and rules that govern a system. A system can be defined by a set of rules, like in a game. In science, we assume that natural systems are governed by a finite set of rules. The search for these basic rules, or fundamental laws as they are called in this case, was the central scientific quest for centuries. It is not easy to determine such laws, but enormous progress has been made nevertheless. Newton's laws defining classical mechanics, or the Maxwell equations of electromagnetism are beautiful examples of fundamental laws in nature. We do not have a theory of the brain on this level, and some have argued that we might never find a simple set of rules explaining brain functions. However, in many scientific disciplines we are beginning to realize that even with a given set of rules we might still not have a sufficient understanding of the underlying systems. This is analogous to the idea that knowing the rules of the card game Bridge is not sufficient to be a good Bridge player.

Rules define a system completely, and it can therefore be argued that all the properties of a system are encoded in the rules. However, we have to realize that even a small set of rules can generate a multitude of behaviours of the systems, which might be difficult to understand from the basic rules due to the presence of emergent properties. A different level of description might then be more appropriate. For example, thermodynamics can describe appropriately the macroscopic behaviour of systems of many weakly interacting particles, even though the systems are governed by other microscopic rules. On the other hand, there are emergent properties in Newtonian systems that are not well described by classical thermodynamics, such as turbulent fluids. A deeper

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understanding of emergent properties is becoming a central topic in the science of the 21st century.

The importance of emergent properties in networks of simple processing elements is not the only extension of traditional information-processing approaches that we think are crucial for understanding brain functions. Another important ingredient is that the brain is an example of an *adaptive system*. In our context, we define adaptation as the ability of a system to adjust its response to external stimuli in dependence of the environment states and the expectations of the system. Humans are a good example of systems that have mastered adaptive abilities and learning. Adaptation and learning is an area that has attracted a lot of interest in the engineering community. The reason for this is that learning systems, which are systems that are able to change their behaviour based on examples in order to solve information-processing demands, have the potential to solve problems where traditional algorithmic methods have not been able to produce sufficient results. Adaptation has two major virtues. One is, as just mentioned, the promise to solve information-processing problems for which explicit algorithms are not yet known. A second virtue is our aim to build systems that can cope with continuously changing environments. A lot of research in the area of neural networks is dedicated to the understanding of learning. Engineering applications of neural networks are not bound by biological plausibility. In contrast to this, we concentrate in this book on biologically plausible learning mechanisms that can help us to comprehend the functionality of the brain.

1.3.2 Levels of analysis

David Marr and Tomaso Poggio realized the usefulness of distinguishing between different levels of description when explaining processes, which Marr later explains beautifully in his posthumously published book *Vision*. One issue that Marr raised in his book was that different people need different kinds of explanations. Explanations of brain functions that are satisfactory for a non-specialist can be insufficient for a physiologist or psychologist who is trying to make sense of specific observations in his or her experiments. A computer engineer may require different types of explanations of brain functions again, to be able to implement specific information-processing algorithms. Marr also stressed the difference between understanding computers and understanding computation. Knowing how bits are represented and transformed in a computer is far from understanding high-level applications such as the World Wide Web.

Besides these important considerations for any explanatory theory, Marr made an important distinction between different levels of analysis. These are summarized in Marr's book as:

- (1) **Computational theory:** *What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?*
- (2) **Representation and algorithm:** *How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?*

(3) **Hardware implementation:** *How can the representation and algorithm be realized physically?*

The most abstract and general level, that of a computational theory, is concerned with what a process is trying to achieve, and what the principal approach to the solution is. In his book, Marr discusses the example of a cash register, which is adding up the price of goods brought to the checkout counter. This explanation answers the question of what the cash register is doing, but we need also to ask why the cash register is doing it (adding the numbers) in this specific way. The answer to this question is that the rules of adding numbers, rather than multiplying them, encapsulates what we think is appropriate for this process; that prices of individual goods should just add up, independent, for example, of the order in which the goods are presented. Marr assigns great importance to this level and explains in his book:

'To phrase the matter in another way, an algorithm is likely to be understood more readily by understanding the nature of the problem being solved than by examining the mechanism (and hardware) in which it is embodied.'

Having a clear theory of what the brain, or specific brain processes, are trying to accomplish can be a powerful research guide and needs to be considered more in neuroscientific research.

The next level of description is concerned with how the computation, specified on the computational level, is realized on an algorithmic level. Marr considered brain research as an information-processing problem, and he clearly was aware of the duality between representing and processing information. That is, in order to process information, this information has first to be represented in a specific form. For example, numbers can be represented in binary form or with Roman numbers. Representations are important since the specific algorithm used for implementing a process usually depends on the representation. Many different representations are possible, and there are usually many algorithms for each type of representation which can accomplish the task. Different representations can drastically influence algorithms, and different algorithms can have quite different properties, such as robustness or efficiency. Theories in computational neuroscience have strongly focused on representations, on which specific algorithms are built. It should also be mentioned that there are examples that use the arguments in the opposite direction, such as deriving representations from constraints on algorithms, such as robustness. Clearly, the brain must use specific representations, and specific algorithms, and it is the goal of computational neuroscience to help find them.¹

The third level of Marr's scheme is concerned with the physical realization of the algorithms and representation. This is the level to which neuronal biophysics and physiology can speak most directly. Again, there might be several possible physical realizations of a specific algorithm, and we want to understand the physical realizations in the brain. This level of explanation does not only back up the level of explanations above, but is also important in its own right for specific interventions. The different levels are weakly related, and inspiration and constraints from different levels of analysis can guide research on other levels.

¹In the definition of this representational and algorithmic level, Marr speaks specifically of input and output representations, and the process of transforming, or mapping, input to output. We will soon discuss why this mapping view is not adequate for brain functions. In a later section of his book, Marr discusses the difficulty in finding invariant features from changing data. This is an example that is difficult in a mapping framework but which can be resolved easily in the computational theory of the brain presented below. But this criticism should not distract from the importance of this level of explanation.

Since this book is a bottom-up approach of the nervous system, algorithmic theories are necessary. As suggested, it is important to have computational theories. This chapter.

1.4 A computational theory

1.4.1 Why a computational theory?

One of the first questions is: all. While this question is asked in many ways, by many people, the answer is: yes. Wolpert studied appropriate movements of stable lives without called the sea squirts through the ocean brain. So, although it seems that animals

If we accept that to move around might want to move of its species and suggestions, we can. Mathematically, in view, we can imagine sophisticated systems organisms. The level of such evolved systems answer to what the

The brain and the possibility of simulation

If individual brains why could it have reason is that in an individual level, which is unknown is essential the adventurous

Since this book is intended to be an introductory guide, we follow mainly a bottom-up approach, through learning first about some physical properties of the nervous system, how information is represented in the brain, and some algorithmic theories of solving specific problems. A broad knowledge of such issues is necessary for any advanced work in neuroscience. However, as Marr suggested, it is important to guide research more specifically through computational theories. We therefore outline in the following section a broad computational theory of the brain to which we return in more detail in the final chapter.

1.4 A computational theory of the brain

1.4.1 Why do we have brains?

One of the first questions we might want to ask is why we have a brain at all. While this question has likely been asked and answered in many different ways, by many people, the ideas of *Daniel Wolpert* are particularly revealing. Wolpert studies sensorimotor control; how brains are able to supervise appropriate movements in a complex sensory world. He notes that plants live stable lives without a brain and he also tells the story of a little sea creature, called the sea squirt, which is hatched with a small nervous system and swims through the ocean until it settles down on some rock, after which it digests its brain. So, although there might be some bias from his sensorimotor perspective, it seems that animals need a brain to move.

If we accept this answer for now, we can ask why creatures with a brain want to move around. We can answer this question by stating that the creature might want to move around to find food and sexual partners to enable survival of its species and to make evolutionary progress. Even without these concrete suggestions, we can think about some objective the creature wants to achieve. Mathematically, we view this as maximizing an objective function. With this view, we can imagine that nervous systems developed into more and more sophisticated systems that help to achieve the evolutionary objectives of moving organisms. The human brain is thereby, arguably, the most developed example of such evolved systems, even inventing machines to travel. Thus, a more formal answer to what the brain is doing might be:

The brain produces goal-directed behaviour to maximize our probability of survival.

If individual brains exist to maximize the organism's survival capacity, then why could it happen that a teenager gets drunk and falls off a cliff? The reason is that maximizing survival probability acts on a population, not on an individual level. The brain implements a specific strategy for each individual, which itself may not be perfect. Furthermore, experimenting with the unknown is essential for finding new, and ultimately better, solutions, so that the adventurous behaviour of a teenager can ultimately help our society.

1.4.2 The anticipating brain

The next question to ask is how the brain achieves the ability to produce specific goal-directed behaviour. Or, in Marr's view of a computational theory, what is the brain specifically doing and why in its specific way. We could again follow the sensorimotor view and state that the ability to produce goal directed movements enables a creature to react to specific environmental situations. The brain helps in this task by analysing sensory representations of the environment and mapping them to appropriate motor actions. It is possible to imagine that very complex mappings can be implemented in neural tissue. Indeed, we argue in Chapter 6 that feedforward networks are universal approximators, meaning that any functional relation between an input vector, representing sensory states, and an output vector, representing motor actions, can be learned by such networks.

The study of sensory-motor mappings has been important in neuroscientific investigations. However, rather than concentrating on fairly direct response functions, our aim in this book is to understand sophisticated human actions enabled by the brain, and ultimately the entire human mind. The central thesis outlined here is that sophisticated human thought processes can not be achieved by simple feedforward mapping systems, but require sophisticated feedback systems.

To illustrate the importance of feedback systems further, let us take a concrete example. Imagine that we go into a concert hall to listen to a symphony. Let us consider a brief visual experience when entering the concert hall. There are chairs which certainly look different to chairs at home, though people usually have no problem recognizing them as chairs, even in a very brief moment. One can do this even with very brief image presentations. *Simon Thorpe* has shown that visual stimuli of very short durations, only lasting a few tens of milliseconds, are enough to enable the categorization of some of the content of visual stimuli. How is this possible?

Visual scene analysis is quite challenging from an information-processing perspective. The human eye contains around 130 million photoreceptors, so that a visual snapshot on the retina would have 15 MB (megabytes) of data if we only consider binary responses of each photoreceptor. While this is roughly the magnitude of modern digital cameras, it is puzzling that we can not achieve any comparable scene analysis with modern computer systems, despite the fact that computers are more than a million times faster (a computer can easily compute several billion instructions per second, or one instruction in less than a nanosecond (ns), which is very fast compared to the millisecond (ms) time scale of a neuron). The often-suggested answer, that the brain is a large parallel computer, is not sufficient, since only a small number of steps can be processed in the brain in a reasonable time between sensation and perception/response. For example, a common guide for neural processing time is around 10 ms per synaptic stage, such that only 10 to 100 processing steps can be assumed for a task in which humans form percepts or start reacting to stimuli. The problem is that we don't know an algorithm that can work with so few steps. Even basic steps in traditional image processing, such as segmenting an image to separate objects from the background, can take thousands of computational steps.

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How, then, can the fast perception of the brain be achieved? Our thesis here is that fast perception can only be achieved through foreknowledge, and that the human brain functions as a predictive or *anticipatory memory system*. In short, the idea is that we already have a rich knowledge of our commonly encountered environment, and that we can extrapolate this knowledge quickly to individual circumstances. Thus, the brief version of our computational theory can be outlined as:

The brain is an anticipatory memory system that learns to represent expectations of the world, which can be used to generate goal-directed behaviour.

People know about chairs, and possibly even concert halls, so analysis of the visual scene, when entering the concert hall, becomes tractable. We will see that central to this thesis is some form of top-down processing, on top of bottom-up processing for acquiring environmental evidence. Simple bottom-up systems, as described by feedforward networks, also called *perceptrons*, are insufficient for human-like information-processing. The brain is not like most machines that simply react predictably to external commands, it is a more sophisticated system which can interact with the environment. For this we need top-down processing and an active engagement of the system with the environment, which also make brains dependent on their history and current states.

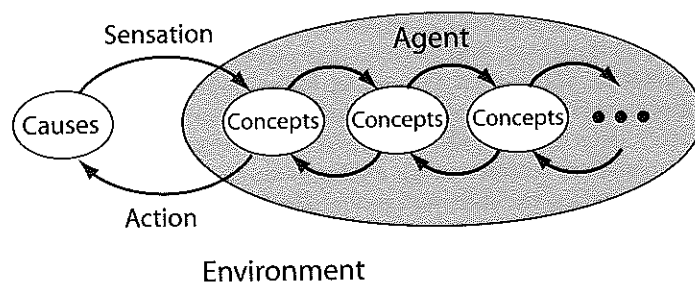


Fig. 1.5 Basic outline of a high-level model of the brain describing a fundamental hypothesis of brain processing as an anticipatory, hierarchical memory system.

An outline of some important components of such an anticipatory system, which we will formalize more fully in Chapter 10, is shown in Fig. 1.5. It is important that the organism acts in an environment through sensations and actions. *Causes* in the environment, such as objects or sequences of events, cause specific sensations to the organism. These sensations activate learned representations in a hierarchical organization within the organism. While the first level or processing is aimed at closely matching peripheral sensations, each additional level is trained to represent increasingly abstract concepts learned from the environment. Such a layered representation is sometimes called a *deep representation*. Concepts are learned, and self-organize, through actions that the organism takes in the environment.

While concepts on different levels can change over time, activations of higher-level concepts will also influence expectations of activating specific lower-level concepts. The anticipatory system tries to match sensations through activating appropriate concepts on different hierarchical levels. This anticipation of

sensations via top-down influences is crucial for the operational capabilities of organisms. Anticipation solves, for example, the time constraints in perception mentioned above, since not all information has to be processed bottom-up to activate the hypothesis of which objects are in the environment. It reduces the complexity of discovering invariant features in the environment, since such knowledge can be learned during development and is part of the expectations of specific sensations.

Also, it is important that organisms act in their environments. For example, objects can be touched, and rotated, to verify whether concepts activated by initial sensations conform with the additional information. This active hypothesis testing can dramatically reduce the learning problem by focusing on relevant data. Organisms can actively explore the environment, which is crucial in forming concepts in the brain. Consequences of the *action-perception loop*, that is, the importance of actions for perception, are only beginning to be explored by the computational research community.

Finally, it is appropriate to introduce and use probabilistic descriptions when talking about many brain processes. We will see that this is simply convenient, on the one hand, but is also required, for the following reasons. Many processes in nature, including brain processes, are inherently noisy. This is reflected by the fact that the outcomes of repeated measurements are always fluctuating. There is considerable debate about the origin of noise in nature, whether noise is a fundamental property, an *irreducible indeterminacy*, or if noise is only based on *epistemic limitations*, such as limitations in the measurement process or the limitations of controlling 'hidden variables'. However, regardless of this philosophical debate, the introduction of the mathematical construct of *random variables*, and the corresponding formal system of *probability theory*, was a major advancement in science. Clearly, the human brain cannot control or accurately determine all the factors in its environment. Thus, epistemic limitations alone are reason enough for the brain to process information probabilistically. Random variables and probability theory are therefore not only a useful description of noisy brain processing, but the fundamental objects described by these concepts may well be fundamental quantities considered by the brain.

Exercises

- (1.1) Give three examples of models and briefly describe what makes them models.
- (1.2) Give four examples of neuroscientific issues at different levels of analysis, as illustrated in Fig. 1.2.
- (1.3) What is the relationship between theoretical and experimental studies in neuroscience?
- (1.4) Make a model of an aeroplane and explain which features are modelled with your solution.

Further

One of the first computational models of the human mind and Sejnowski's work in this area is beyond the level of the book. It contains important rigorous mathematics mentioned in the book and Abbott is a computational model of more general brain processes. It captures important aspects of the help to keep the brain in a state which beautiful models advocated in the book. Doidge (2002) is a book on the consequences of the mind-body problem in psychological studies and applying game theory to the brain.

Further reading

One of the first books that discussed a wide array of computational issues of brain functions was Churchland and Sejnowski (1992). The most advanced book in this area is Dayan and Abbott (2001), which goes beyond the level of the discussions in this book and contains important additional topics, as well as more rigorous mathematical treatment of some of the topics mentioned in this book. While the book by Dayan and Abbott is essential reading for practitioners in computational neuroscience, we also recommend some more general books for a broader audience, which capture important aspects in brain science, and can help to keep the larger picture in mind. Among those books are, foremost, the book by Jeff Hawkins (2004), which beautifully explains the computational theory advocated in this book. The recent book by Norman Doidge (2007) beautifully describes the amazing consequences of brain plasticity. Glimcher (2003) contains a wonderful description of the history of the mind-body philosophy, a great description of physiological studies, as well as some great discussion of applying game theory to neuroscience. Finally, there

is a good collection of articles on Scholarpedia (Izhikevich, 2009), which are a good initial reading to explore important topics in neuroscience.

Patricia S. Churchland and Terrence J. Sejnowski (1992), *The Computational Brain*, MIT Press.

Peter Dayan and Laurence F. Abbott (2001), *Theoretical Neuroscience*, MIT Press.

Jeff Hawkins with Sandra Blakeslee (2004), *On Intelligence*, Henry Holt and Company.

Norman Doidge (2007), *The Brain that Changes Itself: Stories of Personal Triumph from the Frontiers of Brain Science*, James H. Silberman Books.

Paul W. Glimcher (2003), *Decisions, Uncertainty, and the Brain: The Science of Neuroeconomics*, Bradford Books.

Eugene M. Izhikevich (2006), Scholarpedia, <http://www.scholarpedia.org>.

Fundamentals of Computational Neuroscience

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